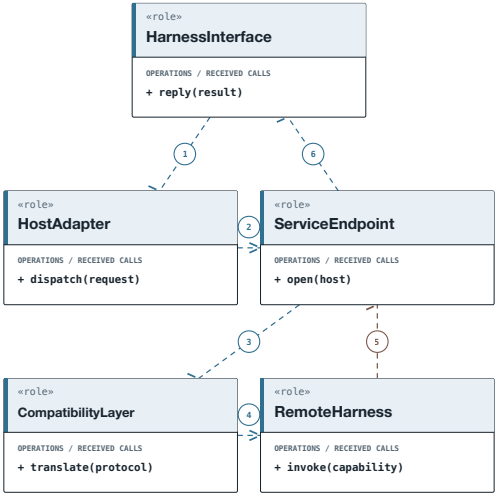
P29 Harness mimicry and service hosting

Primitive / UML CLASS VIEW / ANALYTICAL MODEL

Preserve an interface while changing its implementation or host.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	One harness exposes or consumes an external agent-client or service interface.
TRIGGER	A remote client invokes the adapter or a hosted backend is selected.
STRUCTURE	Protocol adapter, identity, capability mapping, and service boundary.
EFFECT	The external protocol is preserved while semantics and permissions are translated explicitly.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Protocol version and capability mapping are known.
ENSURES	Requests, results, and errors satisfy the compatibility contract.
INVARIANT	Interface compatibility does not erase backend-specific authorization or effects.
VIOLATION	Unsupported capabilities, disconnects, and translation errors are explicit.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 HarnessInterface → HostAdapter dispatch(request)
2 HostAdapter → ServiceEndpoint open(host)
3 ServiceEndpoint → CompatibilityLayer translate(protocol)
4 CompatibilityLayer → RemoteHarness invoke(capability)
5 RemoteHarness ↩ ServiceEndpoint return(result)
6 ServiceEndpoint → HarnessInterface reply(result)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	ACP emulation, server hosting, transport adapter, and backend registry.
TEST_OBLIGATION	Test cross-backend parity, unsupported method, reconnect, and permission mapping.
NEGATIVE_CONTROL	A similar function signature without protocol compatibility is insufficient.